

# Transformer

Why replace recurrence with attention?



Compute across positions in parallel

Recurrent processing



Wait for the preceding state.

$n$  dependent steps

Self-attention



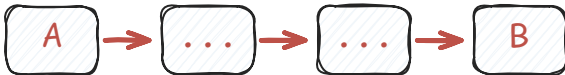
Update all positions together.

⚡ Parallel layer computation



Connect distant tokens directly

Long recurrent path



Pass through intermediate states.

Path grows with sequence length.

One attention hop



Any source pair can interact.

Short paths for long-range context.



Add order; learn multiple relations

Position-aware inputs



Order-aware representations

Multiple attention heads



Learn different relation subspaces

Combine complementary context



Trade-off: dense attention has quadratic token-pair cost.

Training can be parallel; autoregressive generation is sequential.